

Literature Status: Maximal Separated Sets in Super-Reflexive Spaces

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Status. This is a literature-already-answered identification. The source question from Kania–Kochanek, arXiv:1503.08166, Remark 3.7, is explicitly answered by Hajek–Kania–Russo, arXiv:1803.11501.

Source question. Kania and Kochanek study cardinal sizes of subsets of unit spheres whose mutual distances are strictly larger than one. Their Theorem A proves, among other things, that nonseparable quasi-reflexive spaces have uncountable $(1+)$ -separated subsets and that nonseparable super-reflexive spaces have uncountable $(1 + \varepsilon)$ -separated subsets for some $\varepsilon > 0$. In Remark 3.7 they give a reflexive space of density ω_ω for which the supremum in $\mathfrak{eo}(X)$ is not attained at the density of X , and then ask what happens in the super-reflexive case.

Supporting answer. Hajek–Kania–Russo, arXiv:1803.11501, Theorem B, proves the maximal-density super-reflexive statement: if X is super-reflexive, then there exist $\varepsilon > 0$ and a symmetrically $(1 + \varepsilon)$ -separated subset of S_X of cardinality $\text{dens } X$. The introduction explicitly says that this answers the question raised in Kania–Kochanek, Remark 3.7. The same theorem also strengthens the quasi-reflexive part of the source paper by showing that every infinite-dimensional quasi-reflexive space has a symmetrically $(1+)$ -separated subset of S_X with cardinality $\text{dens } X$, and gives a uniform $(1 + \varepsilon)$ version for cardinals of uncountable cofinality.

Scope. This packet does not claim a solution of the fully general nonseparable Kottman problem for arbitrary Banach spaces. It records only the exact super-reflexive Remark 3.7 answer and the associated quasi-reflexive maximal-density strengthening.

References.

- T. Kania and T. Kochanek, *Uncountable sets of unit vectors that are separated by more than 1*, arXiv:1503.08166, *Studia Math.* 232 (2016), 19–44.
- P. Hajek, T. Kania, and A. Russo, *Separated sets and Auerbach systems in Banach spaces*, arXiv:1803.11501, *Trans. Amer. Math. Soc.* 373 (2020), 3113–3145.